

5. **Mesh considerations:** If you use a mesh Wi-Fi system, Wi-Fi 7 can use 6 GHz as a backhaul effectively thanks to the high throughput. This can improve whole-home coverage with less performance loss over hops. So if you have a large home/office, consider Wi-Fi 7 mesh to get strong signals everywhere.
6. **Internet plan:** If your internet connection is, say, 100 Mbps, you won't notice a difference between Wi-Fi 5 vs Wi-Fi 7 for internet tasks – the bottleneck is the ISP. Wi-Fi 7 mainly helps if you have a very fast fiber plan (gigabit or more) or do a lot of local transfers/streaming (like from a home media server or between devices). It's also more about consistency – less likely to drop speed when multiple devices are active.

In summary, Wi-Fi 7 is like moving from a 3-lane highway to a 6-lane superhighway with smarter cars on it. In a congested city (apartment or office), it can alleviate traffic and keep things moving smoothly. Just remember that to enjoy the ride, you need the right car (Wi-Fi 7 laptop) and the road (Wi-Fi 7 router, plus the road being open i.e., 6 GHz allowed). So, when you see a laptop spec “Wi-Fi 6E” vs “Wi-Fi 7”, know that the latter is cutting-edge. Use our checklist to ensure your network environment can take advantage, or plan upgrades accordingly.

7. Battery life and Heat management

Battery life and thermal performance determine the day-to-day experience of using a laptop, yet they're hard to judge from spec sheets alone. Manufacturers might claim “15 hours battery” or use TDP numbers for CPUs, but let's decode how to assess these in the real world:



- **Battery capacity (Wh) and chemistry:** Laptops list battery in Watt-hours (Wh). More Wh generally = longer runtime, all else equal. Small ultrabooks have ~50–60 Wh, bigger ones 70–80 Wh, and some large models 90–100 Wh (which is the typical upper limit for flights). If battery life is a priority, look for a larger Wh number. For instance, the LG Gram 16 packs an 80Wh+ battery in a light chassis, yielding excellent endurance for productivity. That said, capacity is only one factor – the

platform's efficiency is key. Apple's MacBooks, for example, get incredible life from ~50–70 Wh because their ARM-based chips are so efficient. In the Windows world, chips like the Snapdragon X Elite and new Intel Core Ultra aim for better efficiency. A Dell XPS 13 with Snapdragon might claim up to 24 hours on a charge in ideal conditions (local video playback, low brightness) – more realistically it could be 15-18 hours of mixed use, which is still fantastic and far above an older Intel U-series that might manage 8 hours. Always take manufacturer claims with a grain of salt – look for independent tests (e.g., LaptopMag, Notebookcheck) that report web browsing or video playback times. Those give a more realistic figure.

- **Real-world testing cues for battery:** If you can, run a quick test on a new laptop: fully charge it, then do your typical workflow for an hour (web browsing, or streaming a YouTube video). Note the percentage drop. If it drops, say, 10% in an hour doing moderate tasks, that's ~10 hours projected – quite good. If it drops 25% in an hour, that's 4 hours – not great for an unplugged workday. Keep in mind heavy tasks (gaming, rendering) will drain any battery in 1-2 hours, so focus on your average use. Check if the laptop has software for battery profiles – some have a “battery saver” mode that can extend life by reducing performance a bit. Also, observe standby drain – close the lid for a while and see if battery % plummets (some Windows laptops still have issues with standby power draw; modern ones with Modern Standby should do better).
- **Thermals – what to watch:** Laptop cooling is critical to performance and comfort. A powerful CPU/GPU in a thin chassis can run hot and throttle (slow down to cool off). Here's what to check or do:
 - **During a showroom test or initial setup,** put the laptop under a bit of load. For example, open a bunch of browser tabs or stream a 4K video, maybe run a quick free benchmark like Cinebench R24 single then multi. Listen for fan noise – do the fans ramp up aggressively? Is there an annoying whine or is it a smooth whoosh? Some thin laptops can surprise with loud fans even under moderate load. If you intend to push the laptop in workloads, you want fans that balance noise and cooling well.
 - **Feel the chassis:** where does it get warm? It's normal for the area above the keyboard (around the function keys) to act as a heat vent on many laptops. But the keyboard keys where your palms rest shouldn't become uncomfortably hot. Likewise the underside – if you actually use it on your lap, a laptop that hits 45°C bottom surface will be unpleasant. Many gaming laptops run that hot (they're meant to be on a table), whereas ultrabooks usually stay under ~40°C at touch surfaces. If you notice in reviews that “keyboard reaches 50°C under load” – that's a red flag for lap or touch comfort.
 - **Check CPU/GPU temps if possible:** You can install a utility like HWMonitor or use the built-in Xbox Game